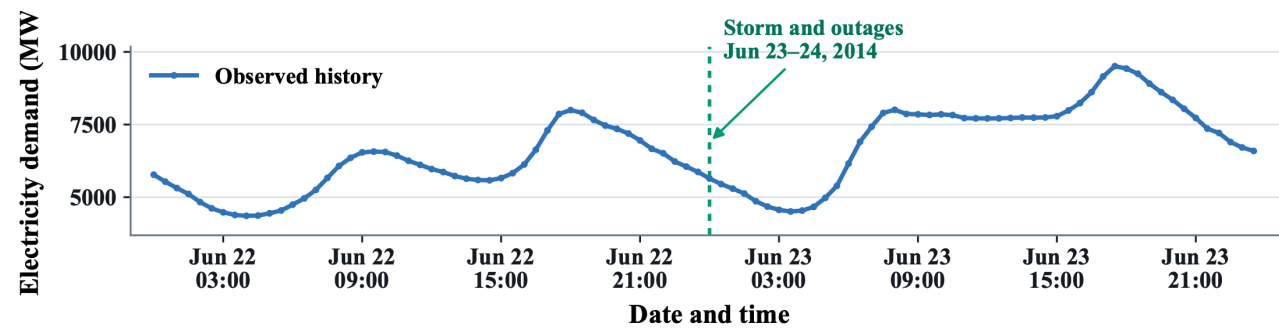


Reasoning Case Study: Why Event Context Can Degrade

NSW electricity demand (MWh) · Forecast horizon: Jun 24, 2014, 00:00–11:30



EVENT

On 23-24 June 2014, destructive winds and blizzards brought down electricity infrastructure and cut power to thousands of homes across Victoria and New South Wales.

Numerical-only

Daily cadence and baseline template

1 “The obvious frequency from observation interval is **daily (48 samples)**. **Forecast horizon 24 samples = half-day**. ...To forecast indices 97-120, previous corresponding indices is 49-72 (**same time previous day**). **Lag 48**. So **seasonal naive** likely $F = [i_{49}..i_{72}] = D2$ early morning

Continuity-adjusted seasonal candidate

2 “Let $\Delta = i_{96} - i_{48} = -0.1911 - (-0.7095) = 0.5184$. Option $F_h = (i_{48} + h?)$...If add Δ to all, $F_h = i_{\{48+h\}} + \Delta$. This is not forecast of same time previous day? It is a “**continuity-adjusted seasonal pattern**”: ..., but **shifted upward** by the current day's 23:30 level difference from the reference 23:30. This accounts for **recent high level**.”

Damping decision

3 “It is like **seasonal naive plus half of boundary anomaly**. It avoids huge drop to -0.87 but still drops to -1.43 low. This may be more balanced. Which one is more “conservative baseline”? Probably **half damping**. I lean to this because recent 23:30 anomaly likely not fully persist; maybe it dissipates; ...Therefore $F = D2 \text{ morning} + 0.26$ may be a good forecast.”

Final forecast rule

4 “Using $\Delta = i_{96} - i_{48} = -0.1911 - (-0.7095) = 0.5184$; **half=0.2592**. $F_j = D2_j + 0.2592$ for $j_{1..24}$ where $D2_j = i_{48+j}$? Let's list D2 early indices 49-72:”

W / Context

Event enters the forecast question

1 “Need predict morning of June 24 after **storm event dates 23-24, destructive winds/cut power**. We must account event may have **residual effect?** Need **infer from history**.”

Weekday baseline, then caution

2 “Thus **Day1 Sunday, Day2 Monday**. Electricity demand NSW higher on weekdays than Sunday, especially morning. So **forecast June24 Tuesday likely similar to Monday**

Competing mechanisms remain unresolved

3 “However likely we should forecast Tuesday based on Monday's profile, perhaps with **slight upward/level continuation** due to Tuesday being another weekday and maybe after **storm restored**. In June winter, blizzard could sed outages could **lower demand below normal**, but we cannot estimate normal from another increase heating demand, but **outages decrease. Hard**.”

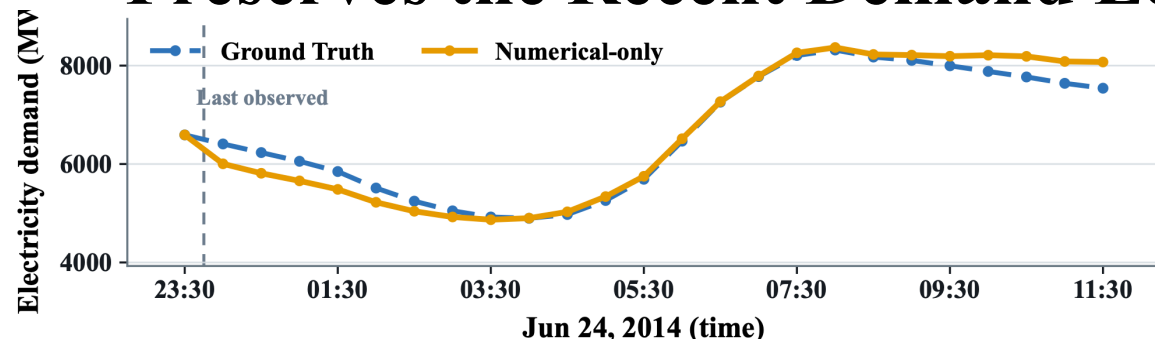
Forecast-day overlap becomes a persistence hypothesis

4 Forecast Tuesday maybe if storm persists (23-24) then outage could also affect Tue; but **24 Jun forecast interval includes morning Jun24**, event says 23-24, so **likely residual event into forecast**. Yet no data for Tue; how revise

Selected late-stage excerpts

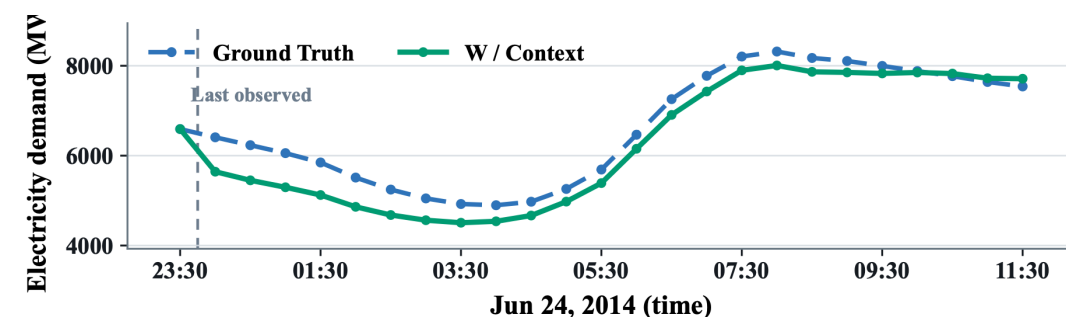
5 “The **residual effect may persist**, so **using Monday's level** (if it reflects event) is good. If we use Monday's actual values, which **already include event impacts**, OK...But no.Let's pick **seasonal naive**.”

Preserves the Recent Demand Level



MAE
196.3

W / Context forecast



MAE
379.1
↑ 93.1